

12.3 Classwork: Polar Form and de Moivre (no calculator) — Markscheme

1. [Maximum mark: 8]

$z = -1 + i\sqrt{3}$, $w = 2 - 2i$, arguments in $-\pi < \theta \leq \pi$.

- (a) Write z in modulus–argument and exponential form. [2]

Solution:

$|z| = \sqrt{(-1)^2 + (\sqrt{3})^2} = 2$; reference angle $\frac{\pi}{3}$, and z lies in the second quadrant, so
 $\arg z = \frac{2\pi}{3}$ **A1**

$$z = 2 \left(\cos \frac{2\pi}{3} + i \sin \frac{2\pi}{3} \right) = 2e^{i\frac{2\pi}{3}} \quad \textbf{A1}$$

A1 for a correct modulus and argument, by any method; an unresolved quadrant ($\pm\frac{\pi}{3}$) earns no mark. A1 for both forms.

- (b) Write w in modulus–argument and exponential form. [2]

Solution:

$|w| = \sqrt{2^2 + (-2)^2} = 2\sqrt{2}$; w lies in the fourth quadrant, so $\arg w = -\frac{\pi}{4}$ **A1**

$$w = 2\sqrt{2} \left(\cos \left(-\frac{\pi}{4} \right) + i \sin \left(-\frac{\pi}{4} \right) \right) = 2\sqrt{2} e^{-i\frac{\pi}{4}} \quad \textbf{A1}$$

Accept $\sqrt{8}$ for $2\sqrt{2}$. An argument of $\frac{7\pi}{4}$ is outside the stated range and earns no mark.

- (c) Find zw and $\frac{z}{w}$ in the form $re^{i\theta}$. [2]

Solution:

$|zw| = 2 \cdot 2\sqrt{2} = 4\sqrt{2}$, $\arg(zw) = \frac{2\pi}{3} - \frac{\pi}{4} = \frac{5\pi}{12}$, so $zw = 4\sqrt{2} e^{i\frac{5\pi}{12}}$ **A1**

$$\left| \frac{z}{w} \right| = \frac{2}{2\sqrt{2}} = \frac{\sqrt{2}}{2}, \arg \frac{z}{w} = \frac{2\pi}{3} + \frac{\pi}{4} = \frac{11\pi}{12}, \text{ so } \frac{z}{w} = \frac{\sqrt{2}}{2} e^{i\frac{11\pi}{12}} \quad \textbf{A1}$$

A1 for each answer; the marks are independent. Accept $\frac{1}{\sqrt{2}}$ or any equivalent exact modulus. FT: award each A1 for a product or quotient formed correctly by the modulus and argument rules from the candidate's own r and θ .

- (d) Plot z , w and zw on the Argand diagram. [2]

Solution:

z at $(-1, \sqrt{3}) \approx (-1, 1.7)$; w at $(2, -2)$ **A1**

zw at $(2\sqrt{3} - 2, 2 + 2\sqrt{3}) \approx (1.5, 5.5)$ **A1**

A1 for z and w plotted and labelled correctly. A1 for zw consistent with modulus ≈ 5.7 and argument $\approx 75^\circ$; accept half-a-grid-square accuracy. FT: use the candidate's (c).

2. [Maximum mark: 7]

(a) Use de Moivre's theorem to find $(1 + i)^8$. [3]

Solution:

$$1 + i = \sqrt{2} \left(\cos \frac{\pi}{4} + i \sin \frac{\pi}{4} \right) \quad \text{M1}$$

$$(1 + i)^8 = \left(\sqrt{2} \right)^8 \left(\cos \frac{8\pi}{4} + i \sin \frac{8\pi}{4} \right) \quad \text{A1}$$

$$(1 + i)^8 = 16(\cos 2\pi + i \sin 2\pi) = 16 \quad \text{A1}$$

M1 for polar or exponential form with a correct modulus and argument. A1 for applying de Moivre's theorem to both the modulus and the argument, however written. A1 for the exact value 16.

Repeated squaring, $(1 + i)^2 = 2i$ then $(2i)^4 = 16$, earns all three marks; note that de Moivre's theorem was asked for rather than withholding marks.

(b) Find the three cube roots of $8i$ in the form $re^{i\theta}$. [3]

Solution:

$$8i = 8e^{i\left(\frac{\pi}{2} + 2k\pi\right)}, k \in \mathbb{Z} \quad \text{M1}$$

$$r = 8^{\frac{1}{3}} = 2 \text{ and } \theta = \frac{\frac{\pi}{2} + 2k\pi}{3}, \text{ so one root is } 2e^{i\frac{\pi}{6}} \quad \text{A1}$$

$$\text{The other roots differ in argument by } \frac{2\pi}{3}: 2e^{i\frac{5\pi}{6}} \text{ and } 2e^{-i\frac{\pi}{2}} \quad \text{A1}$$

M1 for $8i$ in polar or exponential form with the $2k\pi$ (or equivalent multiple-root) allowance shown or used. A1 for the modulus 2 with any one correct argument. A1 for the remaining two roots, each with argument in the required range.

$2e^{i\frac{3\pi}{2}}$ must be converted to $2e^{-i\frac{\pi}{2}}$; the question fixes $-\pi < \theta \leq \pi$. The Cartesian forms $\sqrt{3} + i$, $-\sqrt{3} + i$, $-2i$ may accompany the answer but do not replace the required exponential form.

(c) State the geometric relationship between the three roots. [1]

Solution:

The three roots lie on a circle of radius 2 centred at the origin, equally spaced $\frac{2\pi}{3}$ apart; they are the vertices of an equilateral triangle. **R1**

R1 for any statement carrying both the equal-modulus idea and the equal-spacing idea (equilateral triangle on a circle of radius 2, or arguments differing by 120°). "Equally spaced" alone, with no common modulus or circle, earns no mark.

FT: award R1 for a correct description of the candidate's own three roots from (b).